LNMIIT
The LNM Institute of Information Technology

LNMIIT/B.Tech/IC/2024-25/EVEN/MT121/End Term

The LNM Institute of Information Technology, Jaipur
Department of Mathematics
MT121: Linear Algebra & Complex Analysis
2024-25 EVEN Semester
End Term Examination

Max. Marks: 50 **Date: May 13, 2025** **Duration: 180 Minutes**

Total Questions	Total Marks	CO 1	CO 2	CO 3	CO 4	CO 5
9	50	Q. 1(a), 2(b)	Q. 1(b)	Q. 2(a)	Q.3, Q.7–9	Q.4–6
	CO-PO weightage:	$\frac{4}{50} = 8\%$	$\frac{2}{50} = 4\%$	$\frac{4}{50} = 8\%$	$\frac{23}{50} = 46\%$	$\frac{17}{50} = 34\%$

Instructions: There are total **nine** questions. Your writing should be legible and neat. **No marks will be awarded without proper justification/steps. All parts of a question must be together. Calculator is not allowed.** Maximum marks for a question are shown next to it. Please make an index showing the question number and page number on the front page of your answer sheet in the following format.

Question No.	1	2	3
Page No.			

---

1. (a) If $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is a linear transformation given by $T(x, y, z) = (x - y, x + 2y, z)$, find a basis for the kernel and range of T. [3]

   (b) If $A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$, find a, b, c, d such that $A^5 = aA^3 + bA^2 + cA + dI$. [2]

2. (a) Prove the Pythagoras' Theorem and its converse in an inner product space, namely, two vectors x and y are orthogonal if and only if $||x - y||^2 = ||x||^2 + ||y||^2$. [4]

   (b) Let $S = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. What is the subspace $L(S)$ spanned by S? [1]

3. (a) If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then show that u and v are harmonic in D. [3]

   (b) If $u(x, y) = x^3 - 3x^2y - 3xy^2 + y^3$, then find it's harmonic conjugate and the corresponding analytic function. [2]

4. Find the Laurent series for the function $f(z) = \dfrac{-1}{(z-1)(z-2)}$ in the following domains

   (a) $\{z \in \mathbb{C} : |z| < 1\}$ [2]

   (b) $\{z \in \mathbb{C} : 1 < |z| < 2\}$ [2]

   (c) $\{z \in \mathbb{C} : |z| > 2\}$ [2]

5. (a) Find and classify what kind of singularities the following functions have. Justify your answer.

   (i) $\dfrac{\sinh z}{z^3}$ [1]

   (ii) $\dfrac{1 - \cos z}{z^2}$ [1]

   (iii) $e^{\frac{1}{z}}$ [1]

   (b) Using the theory of contour integrals evaluate the real integral $\displaystyle\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta}$, $a > |b|$. [3]

6. Use Cauchy residue theorem to evaluate the following improper integral $\int_0^\infty \frac{x^2 dx}{(x^2+m^2)(x^2+n^2)}$. [5]

7. (a) Let $\arg z$ denotes the principal argument of z and assume that $-\pi \leq \arg z < \pi$. Let $f(z) = Re(z) + i \arg(z)$. Determine the points in the complex plane where the function is discontinuous. Justify your answer. [3]

   (b) Determine all the zeroes of $\sin z$. Justify your answer. [3]

8. (a) If $f(z)$ is differentiable at every point in $D = \{z \in \mathbb{C} : |z| < 1\}$. Prove that f is analytic in D. [3]

   (b) Let $c \neq 1$ be an nth root of unity. Show that $1 + c + c^2 + ... + c^{n-1} = 0$. [3]

9. (a) Show that the function $f(z)$ is analytic in $\mathbb{C}$ if and only if $f(2z)$ is analytic in $\mathbb{C}$. [3]

   (b) Give an example of a function which is differentiable but not analytic. [3]